

Reboot Sequence Synchronization

Enforcing Strict Boot-Time Execution Order Across dcm-agent and reboot-manager

Change: `reboot-sequence-sync`

Date: 2026-05-06

Status: Proposal — Team Review

Agenda

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1. Problem Statement

User Story

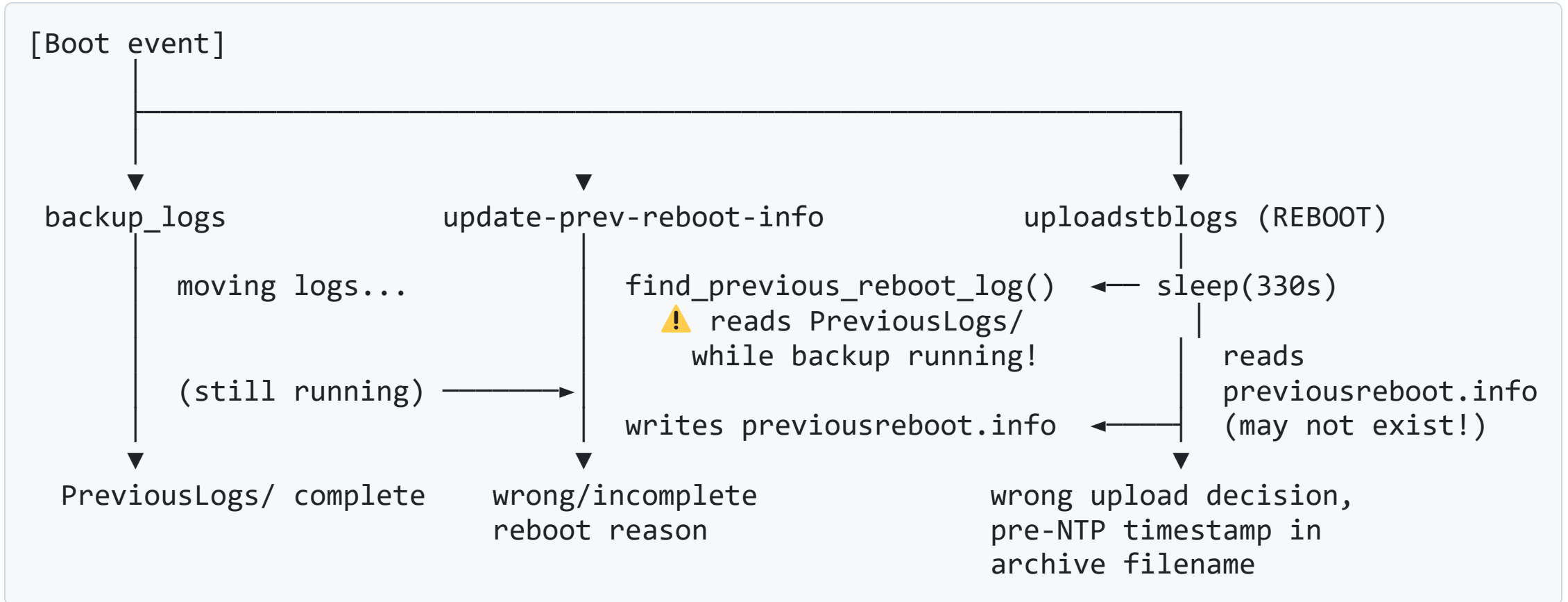
As a device operator, I want log upload after reboot to use the correct reboot reason and NTP-accurate timestamps, so that support teams can reliably correlate uploaded logs with the actual reboot event.

The Gap Today

Three independent boot-time subsystems run **without any ordering guarantee**:

Subsystem	What it does
backup_logs	Moves <code>/opt/logs/*.log</code> → <code>/opt/logs/PreviousLogs/</code>
update-prev-reboot-info	Reads <code>PreviousLogs/</code> to derive reboot reason; writes <code>previousreboot.info</code>

2. Current Architecture — The Race



The 330-second sleep is the only "guard" — and it's not a guard at all.

3. Root Cause — Three Synchronization Gaps

Gap 1 — Backup vs. Reboot Reason Derivation (A → B)

`update-prev-reboot-info` calls `find_previous_reboot_log()` which searches `PreviousLogs/` for the `last_reboot` marker.

Risk: If `backup_logs` is still mid-copy, `last_reboot` may not exist yet, or the directory may be partially populated. The wrong reboot reason gets recorded.

Gap 2 — Reboot Reason vs. Upload (B → C)

`uploadstblogs` reads `previousreboot.info` in `reboot_upload()` to decide whether the reboot was scheduled (DCM-triggered) or unscheduled.

Risk: `previousreboot.info` may not exist yet, or may contain stale data from the prior boot. Wrong data → wrong upload classification.

Gap 3 — NTP Sync vs. Archive Timestamp

`generate_archive_name()` and `add_timestamp_to_files()` both call `time(NULL)`.

Risk: If `uploadstblogs` races ahead of NTP synchronization, archive filenames like `AA-BB_Logs_11-25-25-02-30PM.tgz` carry pre-NTP (often epoch or BIOS) time.

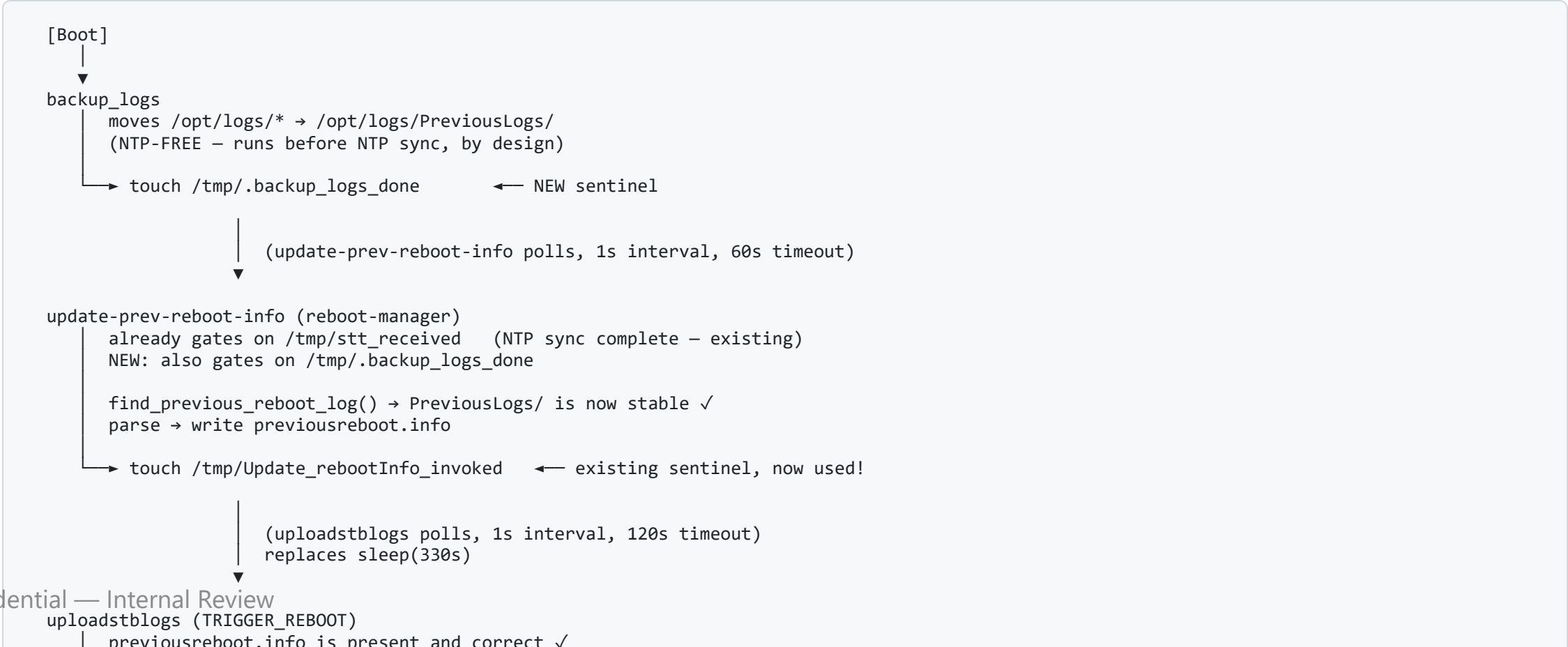
Current workaround: `sleep(330s)` if `uptime < 900s` — crude, wrong direction.

⚠ 330s sleep is:

- Too long on fast/modern devices → needlessly delays uploads
- Too short on slow/loaded devices → NTP may still not be done
- Not event-driven → fundamentally unreliable

4. Proposed Solution — Sentinel Chain

Use sentinel files in `/tmp/` (volatile, cleared on every reboot) to chain the three subsystems into a strict, event-driven execution order.



5. Key Design Decisions

#	Decision	Choice	Rationale
Q1	Poll timeout for uploadstblogs	120 seconds	Shorter than 330s; event-driven so fast devices proceed immediately
Q2	Poll interval	1 second	Matches existing codebase pattern; low CPU overhead
Q3	Timeout behavior (uploadstblogs)	Skip upload entirely	Safe degradation; no partial or mis-classified upload
Q4	Sentinel location	/tmp/	Volatile — auto-cleared on reboot; no stale-sentinel risk
Q5	Timeout behavior (uploadstblogs)	Exit with	Prevents deriving wrong reboot reason

6. What Changes, Where

6.1 dcm-agent — backup_logs/src/backup_logs.c

After `backup_logs_execute()` returns `BACKUP_SUCCESS`:

```
int fd = open("/tmp/.backup_logs_done", O_CREAT | O_WRONLY, 0644);  
if (fd >= 0) close(fd);  
/* Non-fatal if creation fails – downstream will timeout gracefully */
```

New constant in `backup_logs/include/backup_logs.h`:

```
#define BACKUP_LOGS_DONE_FLAG "/tmp/.backup_logs_done"
```

Sentinel written: only on success. Not written on failure.

6.2 reboot-manager — rebootreason_main.c

New poll loop before `find_previous_reboot_log()` :

```
while elapsed < 60s:
    if stat("/tmp/.backup_logs_done") == 0 → proceed
    sleep(1s)

if timeout:
    LOG_ERROR("backup_logs did not complete. Aborting.")
    exit(ERROR_GENERAL)
```

New constants in `update-reboot-info.h` :

```
#define PATH_FLAG_BACKUP_LOGS_DONE    "/tmp/.backup_logs_done"
#define BACKUP_LOGS_POLL_INTERVAL_S  1u
#define BACKUP_LOGS_POLL_TIMEOUT_S   60u
```

`/tmp/.backup_logs_done` is a **cross-repo interface contract**. Path changes must be coordinated.

6.3 dcm-agent — uploadstblogs/src/strategies.c — reboot_setup()

Remove (330-second sleep block):

```
// × DELETE THIS
if (uptime_secs < 900) {
    sleep(330 - uptime_secs/3);
}
```

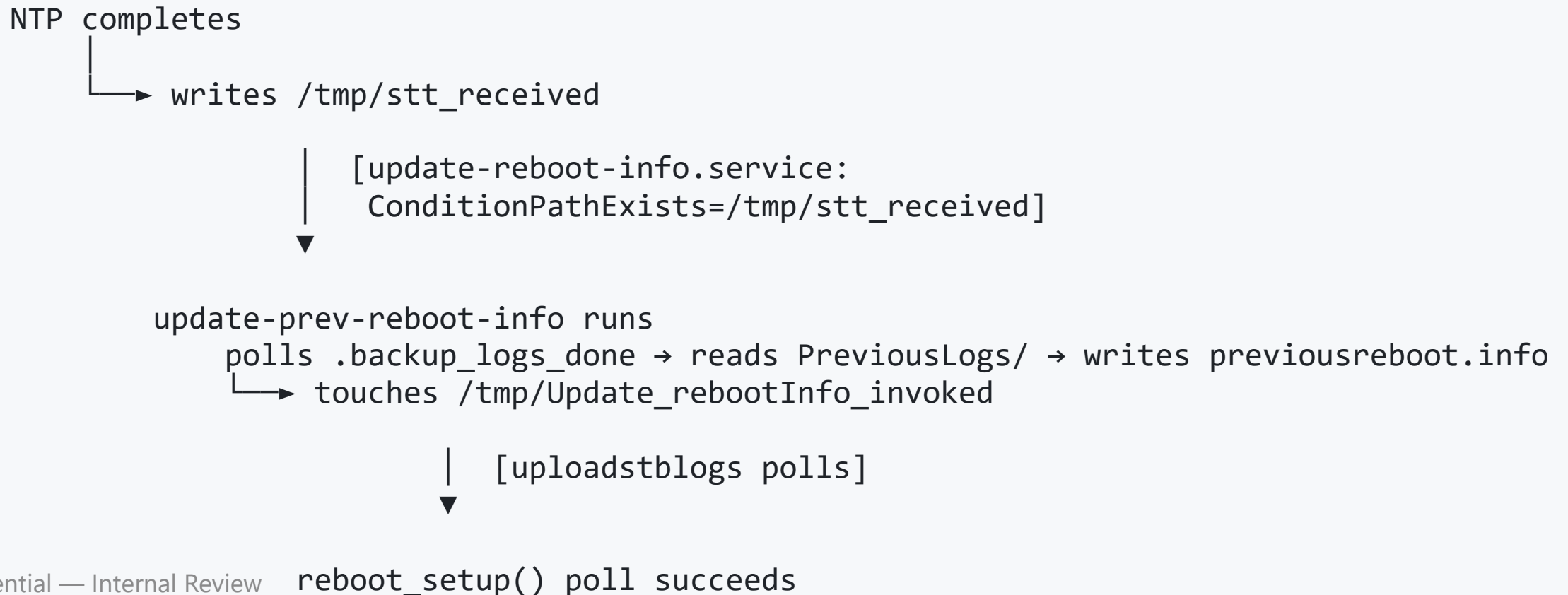
Replace with deterministic sentinel poll:

```
while elapsed < 120s:
    if stat("/tmp/Update_rebootInfo_invoked") == 0 → proceed
    sleep(1s)

if timeout:
    LOG_ERROR("Reboot reason not ready. Skipping reboot upload.")
    return -1 ← strategy handler skips archive/upload/cleanup
```

7. NTP Timestamp Correctness (No Extra Code Needed)

The sentinel chain already **transitively guarantees** NTP sync before any `time(NULL)` call in the upload path:



8. Error & Timeout Behavior

Failure Propagation

backup_logs fails

→ .backup_logs_done NOT written

→ update-prev-reboot-info polls 60s → TIMEOUT → exits ERROR_GENERAL

→ Update_rebootInfo_invoked NOT written

→ uploadstblogs polls 120s → TIMEOUT → skips upload, logs error

→ device continues boot normally

→ next reboot: fresh /tmp/, protocol resets cleanly

Timeout Summary

Subsystem	Polls for	Interval	Timeout	On Timeout
update-prev-reboot-info	/tmp/.backup_logs_done	1s	60s	Exit ERROR_GENERAL
uploadstblogs				Skip reboot

9. Why stat() and Not inotify?

Criterion	inotify	stat() polling
POSIX portability	✗ Linux-only	✓ All target platforms
Embedded resource cost	Higher (kernel watch descriptors)	Minimal
Already in codebase	No	✓ Pattern used in dcm_utils.c
Complexity	Higher (event loop, fd management)	Simple loop
Accuracy	Event-exact	±1s (acceptable)

Decision: stat() polling, consistent with existing codebase patterns.

10. Testing Strategy

Unit Tests

Test	What it verifies	Module
TEST-SYNC-001	Sentinel written on success, absent on failure	backup_logs
TEST-SYNC-002	update-prev-reboot-info proceeds when sentinel present	reboot-manager
TEST-SYNC-003	update-prev-reboot-info exits error after 60s timeout	reboot-manager
TEST-SYNC-004	reboot_setup() proceeds when sentinel present	uploadstblogs

11. Task Breakdown

Groups (parallel where noted)

TASK-A1	Define BACKUP_LOGS_DONE_FLAG in backup_logs.h	[dcm-agent]
TASK-A2	Define PATH_FLAG_BACKUP_LOGS_DONE in update-reboot-info.h	[reboot-manager]
TASK-B1	Write sentinel in backup_logs_main()	[dcm-agent]
TASK-C1	Add poll_for_sentinel() helper	[reboot-manager]
TASK-C2	Gate find_previous_reboot_log() on sentinel	[reboot-manager]
TASK-D1	Add REBOOT_POLL_TIMEOUT_S constant	[dcm-agent]
TASK-D2	Replace sleep(330) with sentinel poll	[dcm-agent]
TASK-E1	Update dcm-agent/docs/DEPENDENCIES.md	[dcm-agent]
TASK-E2	Add reboot-manager/docs/DEPENDENCIES.md entry	[reboot-manager]
TASK-F1	Unit tests – backup_logs sentinel	[dcm-agent]
TASK-F2	Unit tests – reboot-manager poll behavior	[reboot-manager]
TASK-F3	Unit tests – uploadstblogs reboot_setup()	[dcm-agent]
TASK-F4	I2 integration test – full sentinel chain E2E	[dcm-agent]

12. What We Are NOT Changing

To set clear scope expectations:

Area	In scope?	Note
TRIGGER_DCM upload path	✗ No	Only REBOOT trigger has the race
TRIGGER_ONDEMAND path	✗ No	User-triggered; not a boot-sequence concern
Scheduled log upload	✗ No	Different code path entirely
NTP subsystem	✗ No	We consume its signal, don't change it
Systemd service files	✗ No	Existing .service / .path units unchanged
backup_logs copy logic	✗ No	Only sentinel write added at end

13. Summary

Before:

- Three concurrent processes with no ordering
- 330-second sleep as the only "guard"
- Race conditions on `PreviousLogs/` directory and `previousreboot.info`
- Possible pre-NTP archive timestamps

After:

- Deterministic, event-driven execution chain
- `backup_logs` → signals done → `update-prev-reboot-info` → signals done → `uploadstblogs`
- `backup_logs` remains NTP-free
- `uploadstblogs` gets NTP-correct time transitively, automatically

Questions & Discussion

sentinel chain recap:

```
backup_logs
└─> /tmp/.backup_logs_done      (NEW – dcm-agent)
    |
update-prev-reboot-info
└─ polls .backup_logs_done      (NEW – reboot-manager)
  └─> /tmp/Update_rebootInfo_invoked (existing – reboot-manager)
      |
uploadstblogs reboot_setup()
└─ polls Update_rebootInfo_invoked (replaces sleep(330) – dcm-agent)
```

Artifacts: `openspec/changes/reboot-sequence-sync/`

- `proposal.md` — What and why
- `spec.md` — Formal requirements (REQ-SYNC-001 through REQ-SYNC-007)
- `design.md` — Code changes, diagrams, rationale